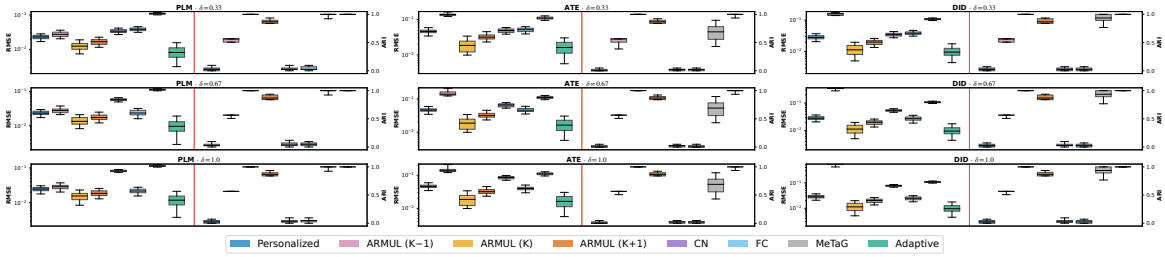

Figure 1: Comparison of estimation accuracy (RMSE on log scale, left panels) and cluster recovery (ARI, right panels) across three models, PLM, ATE, DID, under increasing cluster separation levels $\delta \in \{1/3, 2/3, 1\}$. Each panel shows boxplots over 100 simulations for estimators: Personalized, ARMUL with $K-1$, K, and $K+1$ clusters, CN, FC, MeTaG, and the adaptive estimator. Adaptive method achieves the lowest RMSE and near-perfect ARI, demonstrating accurate estimation and robust recovery of the true latent grouping structure.

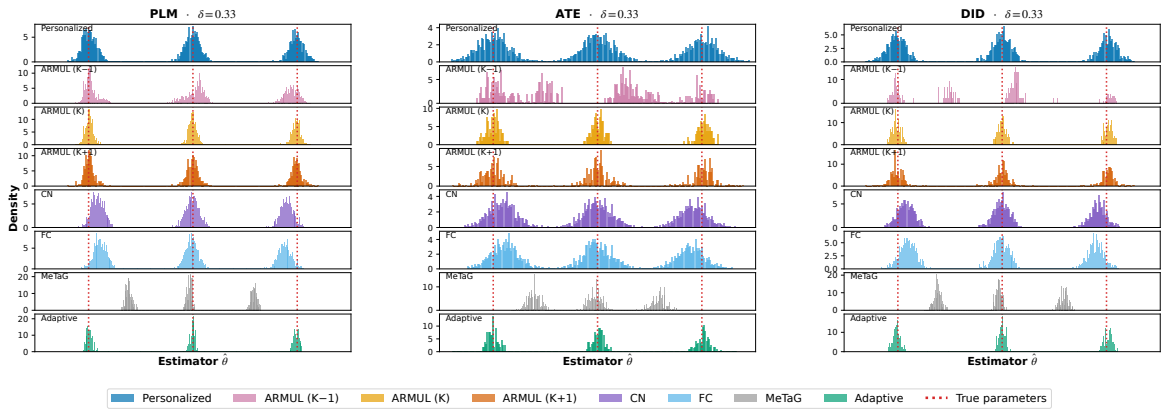

Figure 2: Distribution of task-specific estimators $\hat{\theta}_j$ across three models PLM, ATE, DID, at separation level $\delta = 1/3$. Each column corresponds to one model type, and each row to an estimator: Personalized, ARMUL with $K-1$, K, and $K+1$ clusters, CN, FC, MeTaG, and the adaptive estimator. Red dotted lines mark the true parameters $\beta_1^* = -1/3$, $\beta_2^* = 0$, and $\beta_3^* = 1/3$. The adaptive estimator exhibits the tightest concentration around the true parameters, indicating superior estimation accuracy and stability across models.

Table 1: Weighted RMSE and ARI under PLM, ATE, and DID.

	δ	**Per**	**ARMUL(K-1)**	**ARMUL(K)**	**ARMUL(K+1)**	**CN**	**FC**	**MeTaG**	**Ada**
(a) Weighted RMSE									
PLM	1/3	11.75	14.13	6.60	8.64	17.41	19.59	55.44	**4.46**
	2/3	12.15	14.13	7.21	9.11	29.23	12.40	57.01	**5.27**
	1	12.67	14.52	8.01	9.71	41.83	11.22	58.35	**6.25**
ATE	1/3	23.86	70.88	10.02	16.83	25.07	26.25	56.41	**8.82**
	2/3	23.86	76.66	9.80	16.83	33.02	24.03	56.41	**8.82**
	1	23.86	74.09	9.80	16.83	43.13	20.51	56.41	**8.82**
DID	1/3	14.75	77.88	6.14	10.19	17.85	19.77	54.84	**5.27**
	2/3	14.75	164.66	6.14	10.22	27.62	14.11	54.84	**5.27**
	1	14.75	254.27	6.14	10.23	38.93	12.68	54.84	**5.27**
(b) ARI									
PLM	1/3	0.03	0.55	**1.00**	0.88	0.04	0.04	0.98	**1.00**
	2/3	0.04	0.56	**1.00**	0.88	0.04	0.04	0.98	**1.00**
	1	0.03	0.56	**1.00**	0.88	0.04	0.04	0.98	**1.00**
ATE	1/3	0.02	0.52	**1.00**	0.87	0.02	0.02	0.67	0.99
	2/3	0.02	0.55	**1.00**	0.87	0.02	0.02	0.67	0.99
	1	0.02	0.55	**1.00**	0.87	0.02	0.02	0.67	0.99
DID	1/3	0.03	0.54	**1.00**	0.88	0.03	0.03	0.93	**1.00**
	2/3	0.03	0.56	**1.00**	0.88	0.03	0.03	0.93	**1.00**
	1	0.03	0.56	**1.00**	0.88	0.03	0.03	0.93	**1.00**

Method	MSE	SD (over 10 monte carlo)
Ada	**0.477164**	**0.073777**
Per	0.497458	0.079283
ARMUL(K=2)	0.737223	0.723128
ARMUL(K=3)	0.645249	0.638903
ARMUL(K=4)	0.644739	0.649222

Table 2: Comparison of methods in terms of MSE on a held-out (10%) dataset for CBECS data.

Method	RMSE	SD (over 10 monte carlo)
Per	**0.818051**	0.016061
Ada	0.823930	0.017656
ARMUL(K=2)	0.878603	**0.035594**
ARMUL(K=3)	0.913522	0.067865
ARMUL(K=4)	0.962894	0.083113

Table 3: Comparison of methods in terms of MSE on a held-out (10%) dataset for RECS data. Although personalized MSE is slightly better than our method, our method is better than ARMUL, as we are using a more flexible partial linear model compared to the linear model in ARMUL.